Education and Median Household Income Across U.S. Counties: Evidence from the 2024 ACS

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Repository: https://github.com/yiyangchen2020-afk/-econ3916-midterm-project

Notebook:https://colab.research.google.com/drive/1DqySXnN4DULEsmebReKhqWkbEY7iqQvb?usp=sharing

Question	How is the share of adults aged 25 or older with a bachelor's degree or higher related to median household income across U.S. counties in 2024, controlling for state differences and the share of people aged 65 or older?
Hypothesis (optional)	I expect a positive relationship between education and median household income. Counties with a higher share of adults holding a bachelor's degree or higher may have higher household income because higher education can provide access to better-paying jobs.
Headline finding	Education is positively related to median household income across U.S. counties. In Model 3, counties with a one percentage point higher education share tend to have about 1.43% higher median household income, controlling for age and state. The 95% confidence interval is 1.35% to 1.50%, based on 3,143 observations. This relationship does not show causation.
Data	The data come from the U.S. Census Bureau's 2024 ACS 5-Year Data Profiles. I downloaded the data on October 26, 2026. Each row represents one U.S. county or county equivalent. The final sample contains 3,143 observations after excluding Puerto Rico and one missing income value.
Method	I use OLS regression to examine the relationship between education and log median household income. Model 1 includes education only. Model 2 adds state dummy variables, and Model 3 further adds the share of people aged 65 or older. All three models use HC3 robust standard errors.

1. The Question and the Data (20 pts, about 1.5 pages)

This study examines the relationship between education and median household income across U.S. counties. Education may affect people's job opportunities and earnings, while income differences reflect economic conditions across counties. This question matters to researchers and policymakers who want to better understand these differences. Education levels differ across U.S. counties. Median household income also varies across counties. Counties with more college-educated adults may have higher median household income. However, other county characteristics may affect both education and income. This relationship does not imply causation.

The data come from the U.S. Census Bureau's 2024 ACS 5-Year Data Profiles. The data were downloaded on October 26, 2026. The original dataset contains 3,222 counties and county equivalents, including 78 in Puerto Rico. These 78 observations are excluded because the analysis focuses on the 50 U.S. states and Washington, D.C. Median household income is missing for De Baca County, New Mexico. After excluding Puerto Rico and this missing value, 3,143 observations remain. I treat the missing value as missing completely at random (MCAR). However, this assumption cannot be verified from one missing observation.

Median household income is measured in dollars. Education is measured as the percentage of adults aged 25 or older with a bachelor's degree or higher. The unemployment rate is also measured as a percentage. The share of people aged 65 or older is calculated by dividing the population aged 65 or older by the total population and multiplying by 100. All variables are measured at the county level and use the data from 2024 ACS dataset.

The ACS provides survey-based estimates rather than exact values for every county. The 2024 ACS 5-Year Data Profiles use data collected over five years. Median household income measures income at the household level, and education measures the share of adults aged 25 or older with a college degree. These variables describe counties rather than individual people. Therefore, the results cannot show whether a person with a college degree earns more than a person without one.

Table 1. Summary statistics

Variable (units)	Mean	Median	SD	Min	Max	N
Median household income ($)	**67,976.43**	**65,681**	**17,774.48**	**22,188**	**181,765**	**3,143**
Bachelor's degree or higher, age 25+ (%)	**24.69**	**22.20**	**10.29**	**0.00**	**80.50**	**3,143**
Unemployment rate (%)	**4.66**	**4.40**	**2.41**	**0.00**	**28.60**	**3,143**
Population aged 65+ (%)	**20.44**	**20.00**	**5.08**	**4.47**	**81.82**	**3,143**

Note: All variables are from the U.S. Census Bureau's 2024 ACS 5-Year Data Profiles. The data were downloaded on October 26, 2026.

2. Descriptive Evidence (20 pts, about 1.5 pages)

Median household income is right-skewed across U.S. counties. The mean is $67,976.43, while the median is $65,681. The standard deviation is $17,774.48. Median household income ranges from $22,188 to $181,765. The mean is higher than the median, which is consistent with the right-skewed distribution. A small number of counties have much higher median household incomes than most other counties.

The skewness is 1.2538. Counties with a higher share of adults holding a bachelor's degree or higher tend to have higher median household income. However, this relationship does not imply causation. I use log median household income in the regression because the log transformation reduces skewness to 0.1640.

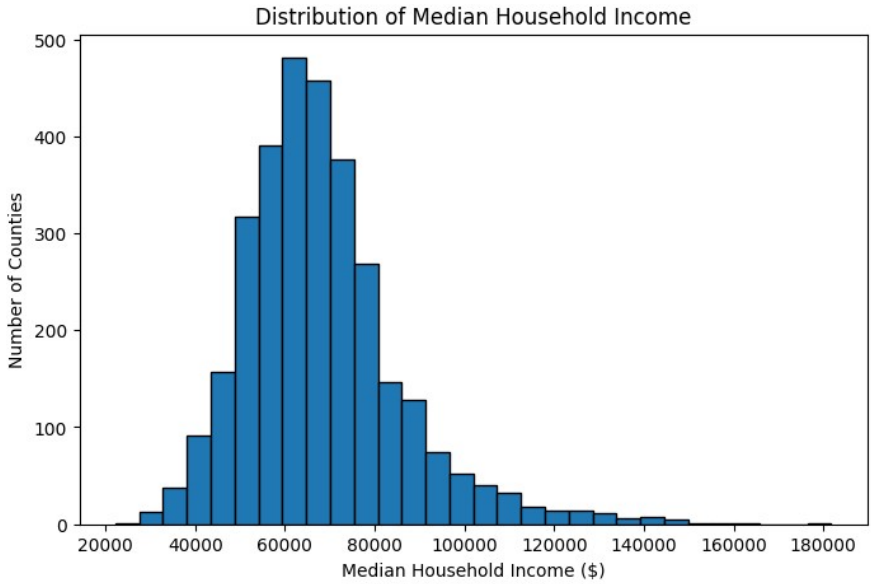


Figure 1. Median household income is right-skewed. Most counties have lower or middle income levels, while a smaller number of counties have much higher incomes. Source: U.S. Census Bureau, 2024 ACS 5-Year Data Profiles.

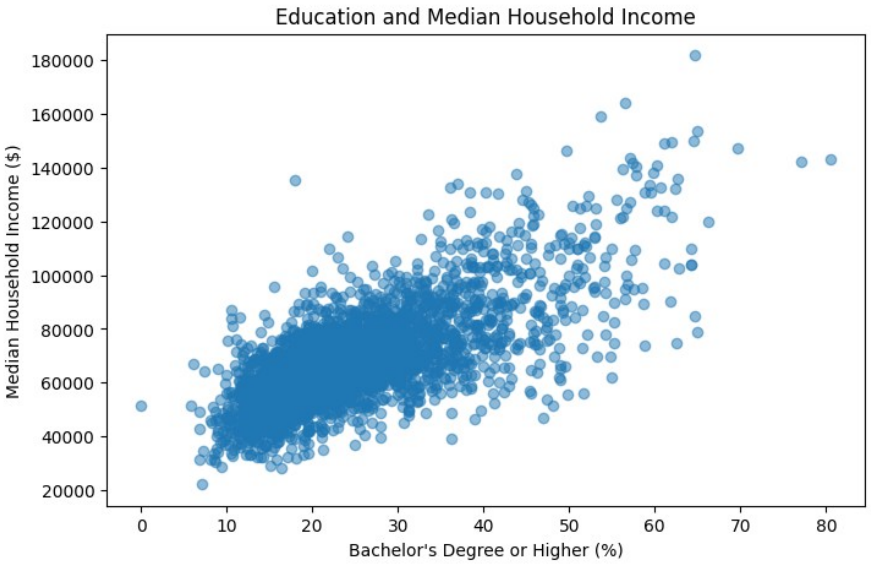


Figure 2. Counties with a higher share of adults holding a bachelor's degree or higher tend to have higher median household income. However, the relationship does not imply causation. Source: U.S. Census Bureau, 2024 ACS 5-Year Data Profiles.

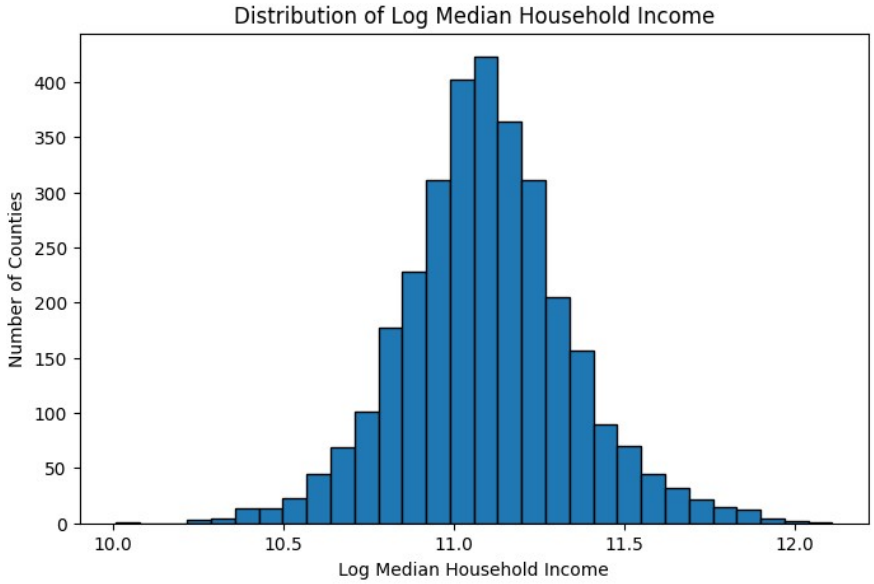


Figure 3. The distribution of log median household income is less right-skewed than the original income distribution. The log transformation reduces skewness from 1.2538 to 0.1640. Source: U.S. Census Bureau, 2024 ACS 5-Year Data Profiles.

3. The Model (25 pts, about 2 pages)

I use OLS regression to examine the relationship between education and log median household income across U.S. counties. Model 1 includes education only. Model 2 adds state dummy variables to control for differences between states. Model 3 further adds the share of people aged 65 or older. All three models use HC3 robust standard errors.

The final regression model is:

$$\log(Income_i) = \beta_0 + \beta_1 Education_i + \beta_2 Age65_i + \sum_{s=2}^{51} \gamma_s State_{is} + u_i$$

Education is positively related to median household income in all three models. The education coefficients are 0.0168, 0.0150, and 0.0143, respectively. The corresponding R-squared values are 0.480, 0.605, and 0.624. In Model 1, counties with a one percentage point higher education share tend to have about 1.68% higher median household income. The education coefficient decreases to 0.0150 after adding state dummy variables and to 0.0143 after adding age. The R-squared increases from 0.480 in Model 1 to 0.605 in Model 2. It increases further to 0.624 in Model 3. This means that the models explain more of the variation in log median household income after adding state and age controls.

In Model 3, education is positively related to median household income. A one percentage point higher education share is related to about 1.43% higher median household income, holding age and state constant. The 95% confidence interval is 1.35% to 1.50%. The share of people aged 65 or older has a negative relationship with median household income. A one percentage point higher share is related to about 0.73% lower income, with a 95% confidence interval of -0.89% to -0.57%. These results do not show causation.

Age is included in Model 3 because it may affect both education and income. State dummy variables are included to control for differences between states. However, they cannot fully account for differences within each state, such as local industries and living costs. Unemployment is not included because education may affect unemployment, which may also affect income. Therefore, unemployment may not be a suitable control variable.

The model compares counties with different education levels while holding age and state constant. However, other county characteristics may still affect the results. The data cover one period rather than changes over time. Therefore, the education coefficient reflects a conditional relationship rather than a causal effect.

4. Diagnostics (20 pts, about 1.5 pages)

Figure A1 shows the residuals against fitted log income. Figure A2 shows the residuals against education. Neither plot shows a clear curved pattern. However, the spread of the residuals is not constant across all fitted values. This suggests that heteroscedasticity may be present. The Breusch-Pagan test checks whether the error variance is constant. The p-value is below 0.001, so the null hypothesis of constant variance is rejected. This supports the evidence of heteroscedasticity. HC3 robust standard errors are used in all three models to make the statistical inference more reliable.

The VIF values for education and age are 1.33 and 1.16, respectively. Both values are low, suggesting that multicollinearity is not severe for these variables. Cook's distance identifies 157 observations above the 4/n cutoff of about 0.0013. One observation in Washington, D.C. has a leverage value of 1, so its Cook's distance is undefined. After removing the 10 most influential counties, the education coefficient changes from 0.0143 to 0.0144. The change is small, suggesting that the main result is not driven by these observations.

5. Threats and Conclusion (10 pts, about 1 page)

The results may be influenced by missing variables such as local industries, living costs, and urban-rural differences. High-income counties may also attract more educated people. The unemployment rate may not be a suitable control variable because education may affect unemployment, which may then affect income. These factors make it difficult to identify causal relationships. For the education coefficient to have a causal interpretation, other factors affecting both education and income would need to be controlled for. This is difficult with the available county-level data. The model includes age and state controls, but it does not account for all differences between counties.

Therefore, the education coefficient reflects a conditional association rather than a causal effect.

The results show a positive relationship between education and median household income across U.S. counties. In Model 3, counties with a one percentage point higher education share tend to have about 1.43% higher median household income, holding age and state constant. After removing the 10 most influential counties, the result remains similar. However, omitted variables and reverse causality limit the interpretation of the results. With more time, I would include county population density to better account for urban-rural differences. These results do not establish causation.

Appendix (does not count toward the 6 to 8 pages)

Table A1. Log Median Household Income Regressed on Education and Controls

Variable	(1) Education Only	(2) + State Dummies	(3) + Age65
Education	0.0168	0.0150	0.0143
HC3 Standard Error	(0.0004)	(0.0004)	(0.0004)
Age65	—	—	-0.0073
Standard Error	—	—	(0.0008)
Constant	10.6819	10.5990	10.7565
Standard Error	(0.0095)	(0.0237)	(0.0284)
State Dummies	No	Yes	Yes
N	3,143	3,143	3,143
R^2	0.480	0.605	0.624

Note: The dependent variable is log median household income. HC3 robust standard errors are reported in parentheses. Model 1 includes education only. Model 2 adds state dummy variables. Model 3 further adds the share of people aged 65 or older. Source: U.S. Census Bureau, 2024 ACS 5-Year Data Profiles.

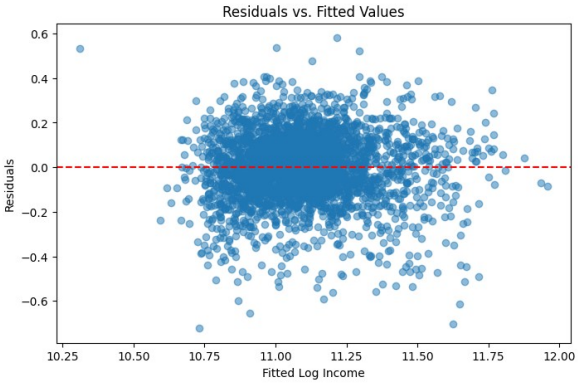


Figure A1. Residuals show no clear curved pattern against fitted values. Source: Author's calculations using the 2024 ACS 5-Year Data Profiles.

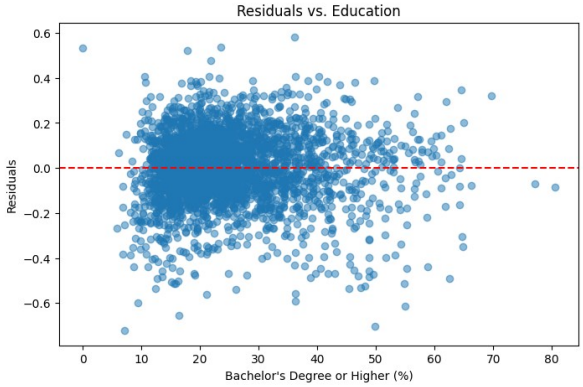


Figure A2. Residuals show no clear curved pattern against education. Source: Author's calculations using the 2024 ACS 5-Year Data Profiles.